

AI Engineer로 취업하려면?



Contents

01

AI Engineer 커리어 소개

AI Engineer 직군의 커리어 및 경력사항을 예시를 통해서 이해합니다.

02

AI Engineer 취업 뽀개기

AI Engineer 직군으로 취업 시 유용한 팁들을 알아갑니다.

03

AI Engineer 취업 후 지속가능한 성장

AI Engineer 직군으로 취업 후 지속가능한 성장을 하는 방법을 알아갑니다.

AI Engineer 커리어 소개

AI Engineer

소위 말하는 연구개발 (R&D) 직종 for AI
필요에 따라 새로운 방법을 고안하기도, 기존의 방법론을 적용하기도

주요 업무

[1]

- LLM에 필요한 연구 및 개발
- 고객의 문제를 이해하고, 그를 가장 효과적으로 해결하기 위한 가설 제시 및 검증을 수행
- 제품의 요구 사항을 이해하고 이를 개발 목표 및 연구 과제로 변환
- 모델 성능에 대한 객관적인 메트릭 정의 및 측정

[1] <https://careers.upstage.ai/o/ai-research-engineer-llm>

AI Engineer 직군의 자격요건/우대사항

AI Engineer 직군에서는 주로 어떤 사항들을 중요하게 여길까?

Requirements

필수 사항

- Computer Science, AI 및 Engineering 등 관련 분야의 학사 학위 또는 동등한 실무 경험
- AI 문제를 정의하고 해결하는 능력
- 머신 러닝, 딥 러닝 및 자연어 처리 분야의 연구 경험
- 주요 기계 학습 프레임워크에 대한 깊은 이해(예: TensorFlow, PyTorch)
- NLP, LLM 등 관심 AI 도메인의 최신 동향에 대한 이해
- 다양한 협업 환경에서 일할 수 있는 능력
- 훌륭한 의사 소통 능력

우대 사항

- Computer Science, AI 및 Engineering 등 관련 분야의 박사 학위 취득 후 3년 이상의 실무 경험
- 대규모 상용 ML(LLM, NLP 등) 제품 개발 경험
- AI 및 NLP 관련 International Conference Main Track (e.g. NeurIPS, ICML, ICLR, CVPR, ICCV, ECCV, ACL, EMNLP, AAAI, etc.)에서 1저자 혹은 교신저자로 출판 기록
- 국제 혹은 국내 AI 대회 (Kaggle 등) 상위권 입상
- Generative AI 및 LLM 연구 경험
- 글로벌 고객 대응을 위한 영어 커뮤니케이션 능력

[1]

[1] <https://careers.upstage.ai/o/ai-research-engineer-llm>

AI Engineer 직군의 자격요건/우대사항

AI Engineer 직군에서는 주로 어떤 사항들을 중요하게 여길까?

지원자격

- 관련 분야 석사/박사 학위 소지
- Computer Vision/Deep Learning/Machine Learning 알고리즘 연구/개발 경험 보유
- Python, C/C++, Java 프로그래밍 언어의 숙련된 활용 능력
- PyTorch, Keras, Tensorflow 등 최신 Machine Learning 개발 환경에 숙련된 활용 능력
- AI관련 top-tier 논문 출판 또는 경진대회 입상자 우대 (예: CVPR, ICCV, ECCV, NeurIPS, ICML, ICLR 등)

전공

- 컴퓨터공학, 전산학, 전자공학, 산업공학, 생물정보학

[2]

[2] <https://www.lgresearch.ai/careers/view?seq=201>

AI Engineer 직군의 자격요건/우대사항

“Top-tier” 논문 출판

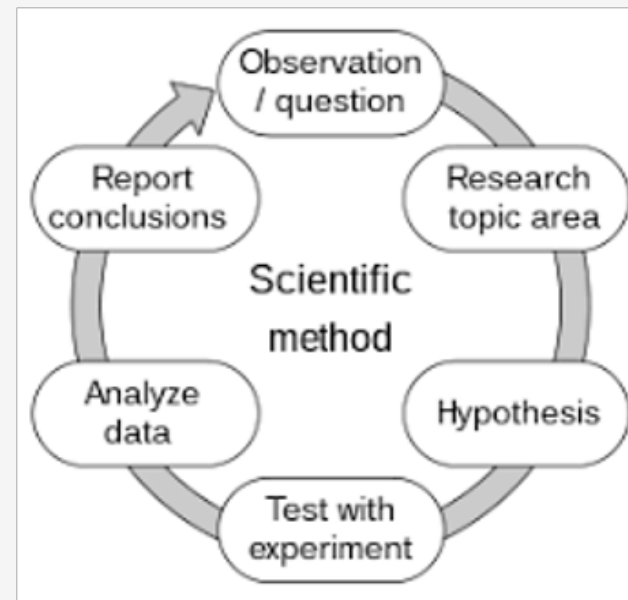
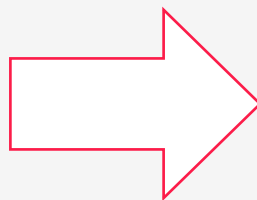


논문 성과를 보는 이유

문제 해결 역량을 보는 데에 가장 적합

논문 = 가설을 뒷받침할 실험적 근거 + 논리적인 글쓰기

추가적인 요구사항



[3]

[3] https://en.wikipedia.org/wiki/Scientific_method

논문 성과를 보는 이유

논문 = 가설을 뒷받침할 실험적 근거 + 논리적인 글쓰기

Model	Size	Type	H6 (Avg.)	ARC	HellaSwag	MMLU	TruthfulQA	Winogrande	GSM8K
SOLAR 10.7B-Instruct	~ 11B	Alignment-tuned	74.20	71.08	88.16	66.21	71.43	83.58	64.75
Qwen 72B	~ 72B	Pretrained	73.60	65.19	85.94	77.37	60.19	82.48	70.43
Mixtral 8x7B-Instruct-v0.1	~ 47B	Instruction-tuned	72.62	70.22	87.63	71.16	64.58	81.37	60.73
Yi 34B-200K	~ 34B	Pretrained	70.81	65.36	85.58	76.06	53.64	82.56	61.64
Yi 34B	~ 34B	Pretrained	69.42	64.59	85.69	76.35	56.23	83.03	50.64
Mixtral 8x7B-v0.1	~ 47B	Pretrained	68.42	66.04	86.49	71.82	46.78	81.93	57.47
Llama 2 70B	~ 70B	Pretrained	67.87	67.32	87.33	69.83	44.92	83.74	54.06
Falcon 180B	~ 180B	Pretrained	67.85	69.45	88.86	70.50	45.47	86.90	45.94
SOLAR 10.7B	~ 11B	Pretrained	66.04	61.95	84.60	65.48	45.04	83.66	55.50
Qwen 14B	~ 14B	Pretrained	65.86	58.28	83.99	67.70	49.43	76.80	58.98
Mistral 7B-Instruct-v0.2	~ 7B	Instruction-tuned	65.71	63.14	84.88	60.78	68.26	77.19	40.03
Yi 34B-Chat	~ 34B	Instruction-tuned	65.32	65.44	84.16	74.90	55.37	80.11	31.92
Mistral 7B	~ 7B	Pretrained	60.97	59.98	83.31	64.16	42.15	78.37	37.83

[4]

[4] <https://arxiv.org/abs/2312.15166>

논문 성과를 보는 이유

논문 = 가설을 뒷받침할 실험적 근거 + 논리적인 글쓰기

SOLAR 10.7B: Scaling Large Language Models with Simple yet Effective Depth Up-Scaling

Dahyun Kim*, Chanjun Park*[†], Sanghoon Kim*[†], Wonsung Lee*[†], Wonho Song*
Yunsu Kim*, Hyeonwoo Kim*, Yungi Kim, Hyeonju Lee, Jihoo Kim
Changbae Ahn, Seonghoon Yang, Sukyung Lee, Hyunbyung Park, Gyoungjin Gim
Mikyong Cha, Hwalsuk Lee[†], Sunghun Kim[†]

Upstage AI, South Korea

{kdahyun, chanjun.park, limerobot, wonsung.lee, hwalsuk.lee, hunkim}@upstage.ai

Abstract

We introduce SOLAR 10.7B, a large language model (LLM) with 10.7 billion parameters, demonstrating superior performance in various natural language processing (NLP) tasks. Inspired by recent efforts to efficiently up-scale LLMs, we present a method for scaling LLMs called depth up-scaling (DUS), which encompasses depthwise scaling and continued pre-training. In contrast to other LLM up-scaling methods that use mixture-of-experts, DUS does not require complex changes to train and inference efficiently. We show experimentally that DUS is simple yet effective in scaling up high-performance LLMs from small ones. Building on the DUS model, we additionally present SOLAR 10.7B-Instruct, a variant fine-tuned for instruction-following capabilities, surpassing Mixtral-8x7B-Instruct. SOLAR 10.7B is publicly available under the Apache 2.0 license, promoting broad access and application in the LLM field¹.

ciently and effectively scale-up LLMs, they often require non-trivial changes to the training and inference framework (Gale et al., 2023), which hinders widespread applicability. Effectively and efficiently scaling up LLMs whilst also retaining the *simplicity* for ease of use is an important problem (Alberts et al., 2023; Fraiwan and Khasawneh, 2023; Sallam et al., 2023; Bahrini et al., 2023).

Inspired by Komatsuzaki et al. (2022), we present depth up-scaling (DUS), an effective and efficient method to up-scale LLMs whilst also remaining straightforward to use. DUS consists of scaling the number of layers in the base model and continually pretraining the scaled model. Unlike (Komatsuzaki et al., 2022), DUS does not scale the model using MoE and rather use a depthwise scaling method analogous to Tan and Le (2019) which is adapted for the LLM architecture. Thus, there are no additional modules or dynamism as with MoE, making DUS immediately compatible with most LLM frameworks.

[4]

[4] <https://arxiv.org/abs/2312.15166>

분야별 Top-tier 논문과 익숙해지기

Google scholar 활용



Metrics

 **Top publications**
Top cited publications over the last five years [Learn more](#)

Publication	h5-index	h5-median
1. Nature	467	707
2. The New England Journal of Medicine	439	876
3. Science	424	665
4. IEEE/CVF Conference on Computer Vision and Pattern Recognition	422	681
5. The Lancet	368	688
6. Nature Communications	349	456
7. Advanced Materials	326	415
8. Cell	316	503
9. Neural Information Processing Systems	309	503
10. International Conference on Learning Representations	303	563

[VIEW ALL](#)

[5]

[5] https://scholar.google.com/citations?view_op=metrics_intro&hl=en

분야별 Top-tier 논문과 익숙해지기

Computer Vision 분야의 탑티어

Categories > Engineering & Computer Science > Computer Vision & Pattern Recognition ▾

h5-index is the h-index for articles published in the last 5 complete years. It is the largest number h such that h articles published in 2018-2022 have at least h citations each. [hide](#)

	Publication	h5-index	h5-median
1.	IEEE/CVF Conference on Computer Vision and Pattern Recognition	422	681
2.	European Conference on Computer Vision	238	390
3.	IEEE/CVF International Conference on Computer Vision	228	366

[5] https://scholar.google.com/citations?view_op=metrics_intro&hl=en

분야별 Top-tier 논문과 익숙해지기

Machine Learning 분야의 탐티어

Categories > Engineering & Computer Science > Artificial Intelligence ▾

	Publication	h5-index	h5-median
1.	Neural Information Processing Systems	309	503
2.	International Conference on Learning Representations	303	563
3.	International Conference on Machine Learning	254	463
4.	AAAI Conference on Artificial Intelligence	212	344

[5]

[5] https://scholar.google.com/citations?view_op=metrics_intro&hl=en

분야별 Top-tier 논문과 익숙해지기

(번외) AAAI

Categories > Engineering & Computer Science > Artificial Intelligence ▾

Publication	h5-index	h5-median
1. Neural Information Processing Systems	309	503
2. International Conference on Learning Representations	303	563
3. International Conference on Machine Learning	254	463
4. AAAI Conference on Artificial Intelligence	212	344

[5]

[5] https://scholar.google.com/citations?view_op=metrics_intro&hl=en

분야별 Top-tier 논문과 익숙해지기

(번외) AAAI

- Three technical tracks (Main Track; AI for Social Impact; Safe, Robust and Responsible AI)
- Two-phase reviewing: two reviews in Phase 1, additional reviews in Phase 2 for papers not rejected in Phase 1. Author response after Phase 2, only for papers not rejected in Phase 1.
- Three kinds of supplementary material may be submitted alongside all papers: (1) technical appendix; (2) multimedia; (3) code and data. Supplementary material deadline is 3 days after the paper submission deadline.
- Additional pages for references only.
- All authors must complete a reproducibility checklist, which facilitates replication of the reported research.
- All authors are expected to be available to review (light load), unless extenuating circumstances apply.
- NeurIPS-23 Fast Track: rejected papers with *final, average* scores of at least 4.9 may be submitted directly into AAAI-24's Phase 2 along with, previous reviews and paper ID.

[6]

[6] <https://aaai.org/aaai-conference/aaai-24-call-for-proposals/>

분야별 Top-tier 논문과 익숙해지기

Natural Language Processing 분야의 탑티어

[5]

Categories > Engineering & Computer Science > Computational Linguistics ▾

	Publication	h5-index	h5-median
1.	Meeting of the Association for Computational Linguistics (ACL)	192	329
2.	Conference on Empirical Methods in Natural Language Processing (EMNLP)	176	302
3.	Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (HLT-NAACL)	133	236

[5] https://scholar.google.com/citations?view_op=metrics_intro&hl=en

분야별 Top-tier 논문과 익숙해지기

CV: CV + ML 탐티어, NLP: NLP + ML 탐티어

Categories > Engineering & Computer Science > Computer Vision & Pattern Recognition ▾

h5-index is the h-index for articles published in the last 5 complete years. It is the largest number h such that h articles published in 2018-2022 have at least h citations each. [hide](#)

	Publication	h5-index	h5-median
1.	IEEE/CVF Conference on Computer Vision and Pattern Recognition	422	681
2.	European Conference on Computer Vision	238	390
3.	IEEE/CVF International Conference on Computer Vision	228	366

Categories > Engineering & Computer Science > Computational Linguistics ▾

	Publication	h5-index	h5-median
1.	Meeting of the Association for Computational Linguistics (ACL)	192	329
2.	Conference on Empirical Methods in Natural Language Processing (EMNLP)	176	302
3.	Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (HLT-NAACL)	133	236

[5]

Categories > Engineering & Computer Science > Artificial Intelligence ▾

	Publication	h5-index	h5-median
1.	Neural Information Processing Systems	309	503
2.	International Conference on Learning Representations	303	563
3.	International Conference on Machine Learning	254	463
4.	AAAI Conference on Artificial Intelligence	212	344

[5] https://scholar.google.com/citations?view_op=metrics_intro&hl=en

AI Engineer 취업을 준비하는 입장에서

논문은 자격증이 아니다

- 연구에 관심이 없는데 순전히 스펙을 위해서 대학원 진학 후 논문 작성을 생각하는 것은 비추
- 연구는 기본적으로 정답이 없는 문제부터 정의하고 해당 문제를 푸는 과정
 - 1~2년 동안 아무런 성과가 없다가 갑자기 실험결과가 잘 나오는 경우가 허다함

AI Engineer 취업을 준비하는 입장에서

문제를 해결한 상황을 많이 만들어라

- 기술을 써보는건 누구나 할 수 있다
 - ex) A 논문을 읽고 해당 기술을 써봤더니 성능이 오른 경험이 있었다
- 기술이 아닌 문제를 해결한 경험을 쌓아라
 - ex) 사람들이 재활용 쓰레기를 잘 버리는지 판별해주는 프로그램을 만들고 싶었어요. 그래서 재활용 쓰레기의 이미지만 보고 잘 분류하기 위해서...

AI Engineer 취업을 준비하는 입장에서

문제에서 출발하면 경험에 깊이가 더해집니다

- 문제부터 출발을 하게 되면 보통은 데이터부터 보게 됩니다
 - 데이터를 일일이 살피는게 귀찮고 보통 남이 미리 수집해놓은 데이터셋을 쓰는게 쉽지만, 실제 현업에서는 데이터부터 구축해야되는 경우 많음

AI Engineer 취업을 준비하는 입장에서

남들이 하기 싫어하는 부분의 전문가가 되어라

- 보통 딥러닝에서 데이터를 만들거나 수집하는 일은 귀찮고 시간이 많이 쓰이지만, 모델을 학습하고, 새로운 기술들을 적용해보는 것은 코드만 살짝 바꾸면 되어서 쉽습니다.
 - 이 때 남들이 기피하는 부분 (데이터 등)에 대한 경험을 쌓으면 강력한 차별점이 될 수 있습니다

AI Engineer 취업을 준비하는 입장에서

특정 문제 종류를 고르는 것은 후순위

- 현재 유행하고 있는 기술들 위주로 공부하는 것은 좋습니다만, 특정 기술에 목 멜 필요는 없다
 - AI 분야 자체가 워낙 빨리 바뀌기 때문에 특정 기술, 문제만 잘하는 것이 아니라 문제 해결 능력 그 자체가 더 중요

AI Engineer 취업을 준비하는 입장에서

논문 많이 읽기

- 결국 AIRE 직군을 뽑는 면접관들 또한 AIRE이기 때문에 해당 분야의 논문을 많이 본 사람일 것이다
 - 해당 분야의 논문을 많이 써봐서 이 사람들의 언어를 아는 것이 아니라면, 논문을 읽어서라도 그들과 말이 통하게 되어야 한다
 - ex) 그건 bias 아닌가요? 최적화가 덜 되어 있네..등

AI Engineer 직군 이력서 살펴보기

예시 이력서 살펴보기

13, Gurodong-ro 28-gil,
Guro-gu, Seoul

Yungi Kim

(+82) 010-3649-4263
eddie@linercorp.com
github.com/eddie-liner

Research Interests

- Large Language Model, Data-Centric AI
- Recommender Systems, Graph Neural Networks

Education

Hanyang University | Seoul, South Korea

M.S. in Computer Science

- Overall GPA: 4.5/4.5
- Advisor: Prof. Sang-Wook Kim & Member of the Big Data Science Laboratory

Mar. 2021 – Feb. 2023

Hanyang University | Seoul, South Korea

B.S. in Industrial Engineering and B.A. in Economics & Finance

- Overall GPA: 4.41/4.5 & Major GPA: 4.49/4.5 & Rank: 1/67 (Summa Cum Laude)

Mar. 2017 – Aug. 2020

Work Experiences

Liner

- Machine Learning Engineer @Research Agent Division
 - End-to-End Development and Release of a Survey Simulator Product
 - End-to-End Development and Release of a Literature Review Product
 - End-to-End Development and Release of a Citation Recommender Product
- Machine Learning Engineer @Agent Cell
 - End-to-End Development of a Search-LLM Specialized for Multiple Downstream Tasks in the Liner Search Product

Mar. 2023 – Feb. 2025

Upstage

- Technical Leader @LLM Evaluation Team
- AI Research Engineer @LLM Data Team
 - The UpsCorpus Maker Project, which automatically generates all of the training data for Upstage LLM development
 - Leading the Development of a Legal-specialized LLM for Law&Company
 - Development of a Math-specialized LLM for QANDA
 - End-to-End Development of a Reward Model for Quality Filtering of Instruction Datasets
 - End-to-End Development of an LLM-based Translator
- AI Research Engineer @RecSys Team
 - The Marshmallow project, which builds a pipeline using deep learning models for real-time inference
 - Development of a recommendation model for Zamface and LotteOn

Mar. 2023 – Feb. 2025

AI Engineer 직군 이력서 살펴보기

예시 이력서 살펴보기

Publications

Conference Papers

* indicates equal contributions

[C13]

LP Data Pipeline: Lightweight, Purpose-driven Data Pipeline for Large Language Models

Yungi Kim* et al.

Empirical Methods in Natural Language Processing (EMNLP 2025)

[C12]

Rethinking KenLM: Good and Bad Model Ensembles for Efficient Text Quality Filtering in Large Web Corpora

Yungi Kim* et al.

Annual Meeting of the Association for Computational Linguistics (ACL 2025)

[C11]

Dataverse: Open-Source ETL (Extract, Transform, Load) Pipeline for Large Language Models

Hyunbyung Park, Sukyung Lee, Gyoungjin Gim, Yungi Kim, Dahyun Kim, Chanjun Park

Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL 2025)

[C10]

Open Ko-LLM Leaderboard2: Bridging Foundational and Practical Evaluation for Korean LLMs

Hyeonwoo Kim, Dahyun Kim, Jihoo Kim, Sukyung Lee, Yungi Kim, Chanjun Park

Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL 2025)

[C9]

Representing the Under-Represented: Cultural and Core Capability Benchmarks for Developing Thai Large Language Models

Dahyun Kim, Sukyung Lee, Yungi Kim, Attapol Rutherford, Chanjun Park

International Conference on Computational Linguistics (COLING 2025)

AI Engineer 직군 이력서 살펴보기

예시 이력서 살펴보기

Projects	
The UpsCorpus Maker Project Project that (i) automates the collection of large web corpora, (ii) performs quality filtering and domain classification efficiently and effectively, and (iii) generates high-quality instructions based on this data, thereby providing all training data for Upstage LLM, named SOLAR, development in a scalable manner	Jun. 2024 – Oct. 2024
Development of a Legal-Specialized LLM for Law&Company Project that develops a Korean legal-specialized LLM that supports attorneys' drafting tasks by successfully performing Graph RAG and Legal QA	May 2024 – Oct. 2024
Development of a Reward Model for Quality Filtering of Instruction Datasets Improved the performance on the reward bench by 3.17% compared to EURUS-RM-7B, which was the highest-performing model available for quality filtering at the time	Apr. 2024 – Jun. 2024
Development of a Math-Specialized LLM for QANDA - Improved the accuracy of a math-specialized LLM up to 7.71% in terms of average mathematical score - Proposed sequential learning approach that enhances problem-solving ability based on logical skills established in the first stage	Oct. 2023 – Dec. 2023
Development of an LLM-based Translator Project that develops an LLM-Based translator for translating LLM training data	Aug. 2023 – Jun. 2024

AI Engineer 취업 뽐내기

AI Engineer 채용 과정

모집 절차 - 전체 온라인으로 진행

- 서류 전형
- 알고리즘 코딩 테스트
- 디버깅 코딩테스트
- 기술 인터뷰 (1차/2차)
- 컬처 인터뷰
- 최종 인터뷰
- 최종 결과 발표

*절차는 상황에 따라 조정될 수 있습니다.

*최종 인터뷰 후, 레퍼런스 체크 절차가 진행될 수 있습니다.

[1]

[1] <https://careers.upstage.ai/o/ai-research-engineer-llm>

서류 전형

이력서 작성이 핵심

42-17 Gucheon-dong, Paik-gu, Seou, Gyeonggi-do Yungi Kim (+82) 910-5649-4265 edkic@upstage.ai github.com/Kyungji																							
Research Interests - Large Language Model, Data-Centric AI - Recommender Systems, Graph Neural Networks																							
Education Hanyang University Seoul, South Korea Mar. 2021 – Feb. 2023 M.S. in Computer Science Overall GPA: 4.5/4.5 Advisor: Prof. Sang-Wook Kim & Member of the Big Data Science Laboratory Hanyang University Seoul, South Korea Mar. 2017 – Aug. 2020 B.S. in Industrial Engineering and B.A. in Economics & Finance Overall GPA: 4.4/4.5 & Major GPA: 4.49/4.5 & Rank: 1/67 (Summa Cum Laude)																							
Work Experiences Upstage Mar. 2023 – Current - Technical Leader @LLM Evaluation Team - AI Research Engineer @LLM Data Team - The UpCorpus Maker Project, which automatically generates all of the training data for Upstage LLM development - Development of Math- and Legal-specialized LLMs for QANDA and Law&Company, respectively - Development of a Reward Model for Quality Filtering of Instruction Datasets - Development of an LLM-based Translator - AI Research Engineer @RecSys Team - The Marshmallow project, which builds a pipeline using deep learning models for real-time inference - Development of a recommendation model for Zamface and LotteOn																							
Publications * indicates equal contributions Conference Papers [C9] <i>Representing the Under-Represented: Cultural and Core Capability Benchmarks for Developing Thai Large Language Models</i> Dahyun Kim, Sukyung Lee, Yungi Kim, Attapol Rutherford, Chanjun Park International Conference on Computational Linguistics (COLING 2025) [C8] <i>sDPO: Don't Use Your Data All at Once</i> Dahyun Kim, Yungi Kim, Wonho Song, Hyeonwoo Kim, Yansu Kim, Sanghoon Kim, Chanjun Park International Conference on Computational Linguistics (COLING 2025) [C7] <i>Evalverse: Unified and Accessible Library for Large Language Model Evaluation</i> Jihoo Kim, Wonho Song, Dahyun Kim, Yansu Kim, Yungi Kim, Chanjun Park Empirical Methods in Natural Language Processing (EMNLP 2024) [C6] <i>SAAS: Solving Ability Amplification Strategy for Enhanced Mathematical Reasoning in Large Language Models</i> Yungi Kim* et al. Empirical Methods in Natural Language Processing (EMNLP 2024) [C5] <i>Open Ko-LLM Leaderboard: Evaluating Large Language Models in Korean with Ko-HS Benchmark</i> Chanjun Park, Hyeonwoo Kim, Dahyun Kim, Seunghwan Cho, Sanghoon Kim, Sukyung Lee, Yungi Kim, Hwaluk Lee Annual Meeting of the Association for Computational Linguistics (ACL 2024) [C4] <i>SOLAR 10.7B: Scaling Large Language Models with Simple yet Effective Depth Up-Sampling</i> Dahyun Kim*, Chanjun Park*, Sanghoon Kim*, Wonsung Lee*, Wonho Song, Yansu Kim, Hyeonwoo Kim, Yungi Kim, Hyeonju Lee, Jihoo Kim, Changbae Ahn, Seonghoon Yang, Sukyung Lee, Hyunhyung Park, Gyeongjin Gim, Miyoung Cha, Hwaluk Lee, Sanghoon Kim Annual Conference of the North American Chapter of the Association for Computational Linguistics (NAACL 2024) [C3] <i>MONET: Modality-Embedding Graph Convolutional Network and Target-Aware Attention for Multimedia Recommendation</i>																							
Yungi Kim* ACM International Conference on Web Search and Data Mining (WSDM 2024) [C2] <i>Is It Enough Just Looking at the Title?: Leveraging Body Text to Enrich Title Words Towards Accurate News Recommendation</i> Yungi Kim* et al. ACM International Conference on Information and Knowledge Management (CIKM 2022) [C1] <i>aiRS: A Large-Scale Recommender System at NAVER News</i> Hongjin Lim*, Yeon-Chang Lee*, Jin-Seo Lee, Sangyu Han, Seunghyeon Kim, Yeonjong Jeong, Changhoon Kim, Jaehun Kim, Sanghoon Han, Solbi Choi, Hanjong Ko, Dokyong Lee, Jaeho Choi, Yungi Kim, Hong-Kyun Bae, Taeho Kim, Jeewon Ahn, Hyun-Seung You, and Sang-Wook Kim IEEE International Conference on Data Engineering (ICDE 2022) Preprints * indicates equal contributions [P6] <i>LP Data Pipeline: Lightweight, Purpose-driven Data Pipeline for Large Language Models</i> Yungi Kim* et al. [P5] <i>Open Ko-LLM Leaderboard2: Bridging Foundational and Practical Evaluation for Korean LLMs</i> Hyeonwoo Kim, Dahyun Kim, Jihoo Kim, Sukyung Lee, Yungi Kim, Chanjun Park [P4] <i>InstaTras: An Instruction-Aware Translation Framework for Non-English Instruction Datasets</i> Yungi Kim, Chanjun Park [P3] <i>1 Trillion Tokens (1TT) Platform: A Novel Framework for Efficient Data Sharing and Compensation in Large Language Models</i> Chanjun Park, Hyeonwoo Kim, Jihoo Kim, Yungi Kim, Dahyun Kim, Sukyung Lee, Seonghoon Yang [P2] <i>Rethinking KenLM: Good and Bad Model Ensembles for Efficient Text Quality Filtering in Large Web Corpora</i> Yungi Kim* et al. [P1] <i>Dataverse: Open-Source ETL (Extract, Transform, Load) Pipeline for Large Language Models</i> Hyunhyung Park, Sukyung Lee, Gyeongjin Gim, Yungi Kim, Dahyun Kim, Chanjun Park																							
Patents Modality-Embedding Graph Convolutional Network and Target-Aware Attention for Accurate Multimedia Recommendation Investor: Sang-Wook Kim, Yungi Kim, Taeri Kim, Application number: 10-2022-0128588, Filing Date: 2022.10.07 Context-aware Attention network with Selection module for Title word representation Investor: Sang-Wook Kim, Taeho Kim, Yungi Kim, Yeon-Chang Lee, Application number: 10-2022-0128555, Filing Date: 2022.10.07																							
Projects The UpCorpus Maker Project Jun. 2024 – Oct. 2024 Project that (i) automates the collection of large web corpora, (ii) performs quality filtering and domain classification efficiently and effectively, and (iii) generates high-quality instructions based on this data, thereby providing all training data for Upstage LLM, named SOLAR, development in a scalable manner Development of a Legal-Specialized LLM for Law&Company May 2024 – Oct. 2024 Project that develops a Korean legal-specialized LLM that supports attorneys' drafting tasks by successfully performing Graph RAG and Legal QA Development of a Reward Model for Quality Filtering of Instruction Datasets Apr. 2024 – Jun. 2024 Improved the performance on the <i>reward bench</i> by 3.17% compared to EURUS-RM-7B, which was the highest-performing model available for quality filtering at the time Development of a Math-Specialized LLM for QANDA Oct. 2023 – Dec. 2023 Improved the accuracy of a math-specialized LLM up to 7.71% in terms of average mathematical score Proposed sequential learning approach that enhances problem-solving ability based on logical skills established in the first stage Development of an LLM-based Translator Aug. 2023 – Jun. 2024 Project that develops an LLM-based translator for translating LLM training data																							
Set up evaluation datasets (Flores, Ko-4F3, and Patent) and metrics (BLEU, CHRF, COMET, BLEURT, and GEMBA) Completed LLM training data and improved the performance of English-to-Korean translation by 3.11% and Korean-to-English translation by 3.94% compared to <i>gpt-4o-0125-preview</i> Released via the Upstage API The Marshmallow Project May 2023 – Jul. 2023 Project that builds pipeline based on deep learning models for real-time inference Improved the accuracy of a recommendation model in the various scenarios up to 36.9% in terms of NDCG@20 and the efficiency of a recommendation pipeline by using Milvus Vector DB and Amazon SageMaker Applied this project to a recommendation model for LotteOn, a leading e-commerce platform Development of a Recommendation Model for Zamface, a E-commerce Platform Mar. 2023 – May. 2023 Improved the accuracy of a recommendation model in the various scenarios up to 129.8% in terms of NDCG@20 and the efficiency of a recommendation pipeline by using Pyspark and Amazon EMR Proposed two stage approach that first generates candidate items and then re-ranks them via LightGBM model Industry Innovation Big Data Platform Contest Apr. 2021 – Jun. 2021 Grand Prize (1st Place) in the Data Analysis Category Competition for multi-label classification using NLP techniques Proposed Sequence-based Attention Neural Network for Sentence Classification, named SANS, that predicts the industry code based on the given business purpose data Awarded by Korea Enterprise Data (KED) Korean Internet New Terms API and Web Dictionary Jul. 2020 – Aug. 2020 3rd Place in the Korea University Course of Data Youth Campus Project for web dictionary on the Korean internet new terms Proposed Korean internet new terms automatic detection model via crawling, sentiment analysis model, and example sentence generation model using NLP techniques <tr><td colspan="4"> Honors & Awards HY-IN Scholarship Mar. 2021 – Feb. 2023 Brain Korea (BK21) Scholarship Mar. 2021 – Feb. 2023 National Science & Technology Scholarship Mar. 2019 – Jul. 2020 Academic Excellence Scholarship Aug. 2017 – Feb. 2019 </td></tr> <tr><td colspan="4"> Certification Advanced Data Analytics Semi-Professional (ADsP) Apr. 2021 SQL Developer (SQLD) Apr. 2021 </td></tr> <tr><td colspan="4"> Teaching Experiences deep daiv., Teacher May 2024 Overview of Large Language Models (Special Lecture) Fast Campus, Teacher in Upstage AI Lab Nov. 2023 – Current PyTorch (Regular Lecture) AI Trend and AI Researcher (Special Lecture) Overview of Large Language Models (Special Lecture) Practice with the Solar API: Generating Synthetic Datasets (Special Lecture) Overview of RecSys & The Concepts of Uninteresting Items (Special Lecture) Introduction to News Recommendation (Special Lecture) Survey on Multimedia Recommendation (Special Lecture) </td></tr> <tr><td colspan="4"> National Institute of Korean Language, Teacher Upstage Data-Centric LLM (Special Lecture) Dec. 2023 </td></tr> <tr><td colspan="4"> Hanyang University, Teaching Assistant Mar. 2021 – Feb. 2023 Creative Programming for Engineers (Spring, Fall 2022) Creative Programming (Spring 2021, 2022) Creative Computing (Fall 2021) </td></tr>				Honors & Awards HY-IN Scholarship Mar. 2021 – Feb. 2023 Brain Korea (BK21) Scholarship Mar. 2021 – Feb. 2023 National Science & Technology Scholarship Mar. 2019 – Jul. 2020 Academic Excellence Scholarship Aug. 2017 – Feb. 2019				Certification Advanced Data Analytics Semi-Professional (ADsP) Apr. 2021 SQL Developer (SQLD) Apr. 2021				Teaching Experiences deep daiv., Teacher May 2024 Overview of Large Language Models (Special Lecture) Fast Campus, Teacher in Upstage AI Lab Nov. 2023 – Current PyTorch (Regular Lecture) AI Trend and AI Researcher (Special Lecture) Overview of Large Language Models (Special Lecture) Practice with the Solar API: Generating Synthetic Datasets (Special Lecture) Overview of RecSys & The Concepts of Uninteresting Items (Special Lecture) Introduction to News Recommendation (Special Lecture) Survey on Multimedia Recommendation (Special Lecture)				National Institute of Korean Language, Teacher Upstage Data-Centric LLM (Special Lecture) Dec. 2023				Hanyang University, Teaching Assistant Mar. 2021 – Feb. 2023 Creative Programming for Engineers (Spring, Fall 2022) Creative Programming (Spring 2021, 2022) Creative Computing (Fall 2021)			
Honors & Awards HY-IN Scholarship Mar. 2021 – Feb. 2023 Brain Korea (BK21) Scholarship Mar. 2021 – Feb. 2023 National Science & Technology Scholarship Mar. 2019 – Jul. 2020 Academic Excellence Scholarship Aug. 2017 – Feb. 2019																							
Certification Advanced Data Analytics Semi-Professional (ADsP) Apr. 2021 SQL Developer (SQLD) Apr. 2021																							
Teaching Experiences deep daiv., Teacher May 2024 Overview of Large Language Models (Special Lecture) Fast Campus, Teacher in Upstage AI Lab Nov. 2023 – Current PyTorch (Regular Lecture) AI Trend and AI Researcher (Special Lecture) Overview of Large Language Models (Special Lecture) Practice with the Solar API: Generating Synthetic Datasets (Special Lecture) Overview of RecSys & The Concepts of Uninteresting Items (Special Lecture) Introduction to News Recommendation (Special Lecture) Survey on Multimedia Recommendation (Special Lecture)																							
National Institute of Korean Language, Teacher Upstage Data-Centric LLM (Special Lecture) Dec. 2023																							
Hanyang University, Teaching Assistant Mar. 2021 – Feb. 2023 Creative Programming for Engineers (Spring, Fall 2022) Creative Programming (Spring 2021, 2022) Creative Computing (Fall 2021)																							

Publications

Conference Papers

* indicates equal contributions

[C9] **Representing the Under-Represented: Cultural and Core Capability Benchmarks for Developing Thai Large Language Models**
Dahyun Kim, Sukyung Lee, Yungi Kim, Attapol Rutherford, Chanjun Park
International Conference on Computational Linguistics (COLING 2025)

[C8] **sDPO: Don't Use Your Data All at Once**
Dahyun Kim, Yungi Kim, Wonho Song, Hyeonwoo Kim, Yansu Kim, Seunghoon Kim, Chanjun Park
International Conference on Computational Linguistics (COLING 2025)

[C7] **Evalverse: Unified and Accessible Library for Large Language Model Evaluation**
Jihoo Kim, Wonho Song, Dahyun Kim, Yansu Kim, Yungi Kim, Chanjun Park
Empirical Methods in Natural Language Processing (EMNLP 2024)

[C6] **SAASe: Solving Ability Amplification Strategy for Enhanced Mathematical Reasoning in Large Language Models**
Yungi Kim* et al.
Empirical Methods in Natural Language Processing (EMNLP 2024)

[C5] **Open Ko-LLM Leaderboard: Evaluating Large Language Models in Korean with Ko-H5 Benchmark**
Chanjun Park, Hyeonwoo Kim, Dahyun Kim, Seunghoon Cho, Seunghoon Yang, Sukyung Lee, Yungi Kim, Hwalbuk Lee

[C4] **SOLAR 10.7B: Scaling Large Language Models with Simple yet Effective Depth Up-Scaling**
Dahyun Kim*, Chanjun Park*, Sanghoon Kim*, Wonsung Lee*, Wonho Song, Yansu Kim, Hyeonwoo Kim, Yungi Kim, Hyeonjin Lee, Jihoo Kim, Changbae Ahn, Seunghoon Yang, Sukyung Lee, Hyunhyung Park, Gyoungjin Gim, MiKyung Cha, Hwalbuk Lee, Sanghae Kim

[C3] **MONET: Modality-Embracing Graph Convolutional Network and Target-Aware Attention for Multimedia Recommendation**

서류 전형

주요 내용부터 작성

- 연락처 -> 관심 분야 -> 학력/경력 사항 -> 논문 & 프로젝트 등 -> 그 외 순서
 - 논문 & 프로젝트에 기재한 연구는 추후 기술면접에서 물어볼 가능성이 매우 큼
 - 이 사람이 실제로 해당 실적에 얼마나 기여했는지 확인하기 위함
 - 본인이 잘 설명할 수 있는 경우만 기재

서류 전형

주요 내용부터 작성

- 뒤로 갈 수록 덜 중요한 내용들
 - 수상 경력
 - 자격증
 - 기타 사항들

• Set up evaluation datasets (Flores, Ko-H3, and Patent) and metrics (BLEU, CHRF, COMET, BLEURT, and GEMBA) for translating LLM training data and improved the performance of English-to-Korean translation by 3.11% and Korean-to-English translation by 3.94% compared to <i>gpt-4-0125-preview</i> • Released via the Upstage API	May 2023 – Jul. 2023
The Marshmallow Project • Project that builds pipeline based on deep learning models for real-time inference • Improved the accuracy of a recommendation model in the various scenarios up to 36.9% in terms of NDCG@20 and the efficiency of a recommendation pipeline by using Milvus Vector DB and Amazon SageMaker • Applied this project to a recommendation model for LotteOn, a leading e-commerce platform	
Development of a Recommendation Model for Zamface, a E-commerce Platform • Improved the accuracy of a recommendation model in the various scenarios up to 129.8% in terms of NDCG@20 and the efficiency of a recommendation pipeline by using Pyspark and Amazon EMR • Proposed two stage approach that first generates candidate items and then re-ranks them via LightGBM model	Mar. 2023 – May. 2023
Industry Innovation Big Data Platform Contest <i>Grand Prize (1st Place)</i> in the Data Analysis Category • Competition for multi-label classification using NLP techniques • Proposed Sequence-based Attention Neural Network for <u>S</u>entence Classification, named SANS, that predicts the industry code based on the given business purpose data • Awarded by Korea Enterprise Data (KED)	Apr. 2021 – Jun. 2021
Korean Internet New Terms API and Web Dictionary <i>3rd Place</i> in the Korea University Course of Data Youth Campus • Project for web dictionary on the korean internet new terms • Proposed korean internet new terms automatic detection model via crawling, sentiment analysis model, and example sentence generation model using NLP techniques	Jul. 2020 – Aug. 2020
Honors & Awards	
• HY-IN Scholarship • Brain Korea (BK21) Scholarship • National Science & Technology Scholarship • Academic Excellence Scholarship	Mar. 2021 – Feb. 2023 Mar. 2021 – Feb. 2023 Mar. 2019 – Jul. 2020 Aug. 2017 – Feb. 2019
Certification	
• Advanced Data Analytics Semi-Professional (ADsP) • SQL Developer (SQLD)	Apr. 2021 Apr. 2021
Teaching Experiences	
deep daiv., Teacher • Overview of Large Language Models (Special Lecture)	May 2024
Fast Campus, Teacher in Upstage AI Lab • PyTorch (Regular Lecture) • AI Trend and AI Researcher (Special Lecture) • Overview of Large Language Models (Special Lecture) • Practice with the Solar API: Generating Synthetic Datasets (Special Lecture) • Overview of RecSys & The Concepts of Uninteresting Items (Special Lecture) • Introduction to News Recommendation (Special Lecture) • Survey on Multimedia Recommendation (Special Lecture)	Nov. 2023 – Current
National Institute of Korean Language, Teacher • Upstage Data-Centric LLM (Special Lecture)	Dec. 2023
Hanyang University, Teaching Assistant • Creative Programming for Engineers (Spring, Fall 2022) • Creative Programming (Spring 2021, 2022) • Creative Computing (Fall 2021)	Mar. 2021 – Feb. 2023

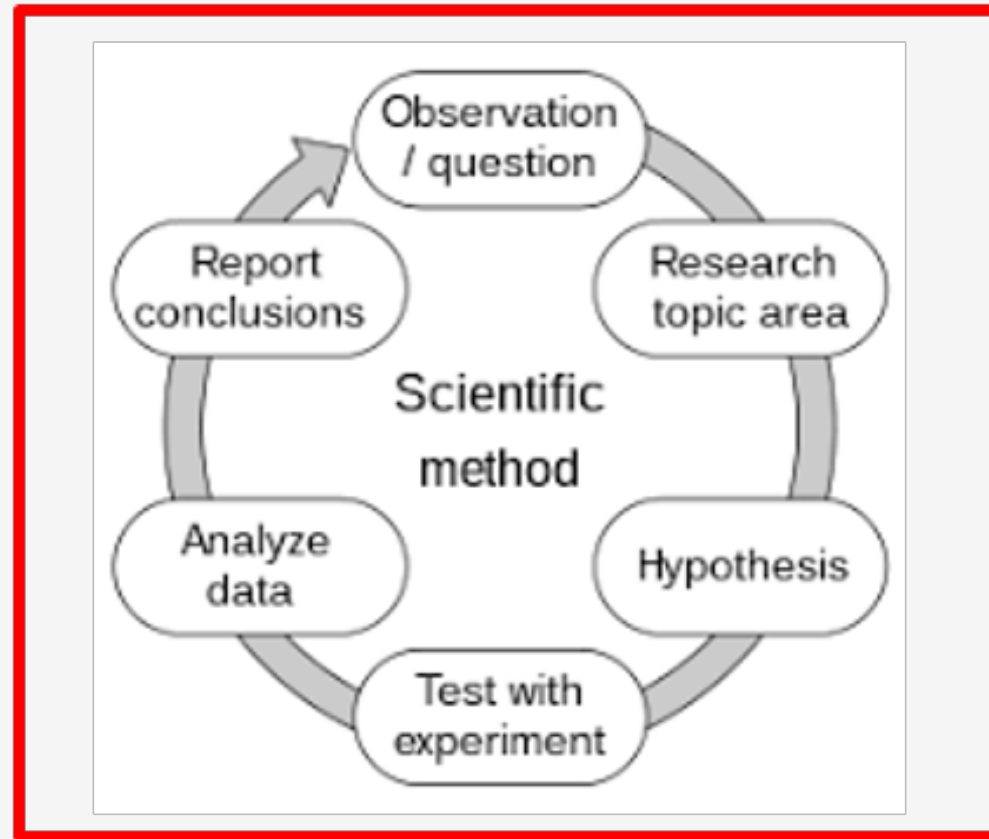
코딩 테스트

일반적인 알고리즘 코딩 테스트

- 다른 개발자들도 보는 일반적인 코딩 테스트로 자신이 없다면 미리 연습 문제 등을 풀어보는 것 추천
- AIRE 직군 자체가 일반적인 알고리즘 지식을 엄청 중요하지는 않아서 커트라인이 상대적으로 낮고 Pass/Fail일 구조일 확률 높음

딥러닝 코딩 테스트

딥러닝 관련 과제를 1~2주 동안 해결하는 테스트



[3]

[3] https://en.wikipedia.org/wiki/Scientific_method

딥러닝 코딩 테스트

딥러닝 관련 과제를 1~2주 동안 해결하는 테스트

- 문제가 주어졌을 때의 문제 해결 능력을 확인하는 테스트
- 데이터셋, 평가 지표 등은 고정으로 주어지고 주어진 기간 내에 성능을 높이는 것이 주 목적
 - 무작정 여러 방법론을 사용X
 - 가설 -> 실험 -> 결과 분석을 하나의 trial로 묶어서 여러 방법을 시도 후 결과 보고에 녹여내는 것이 중요

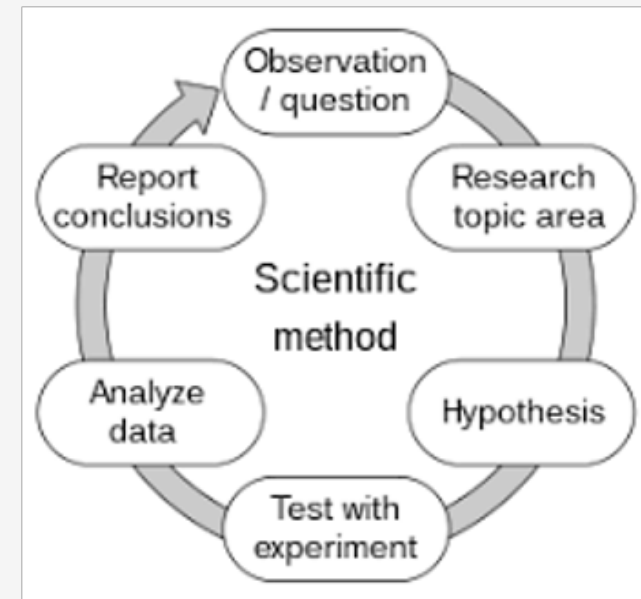
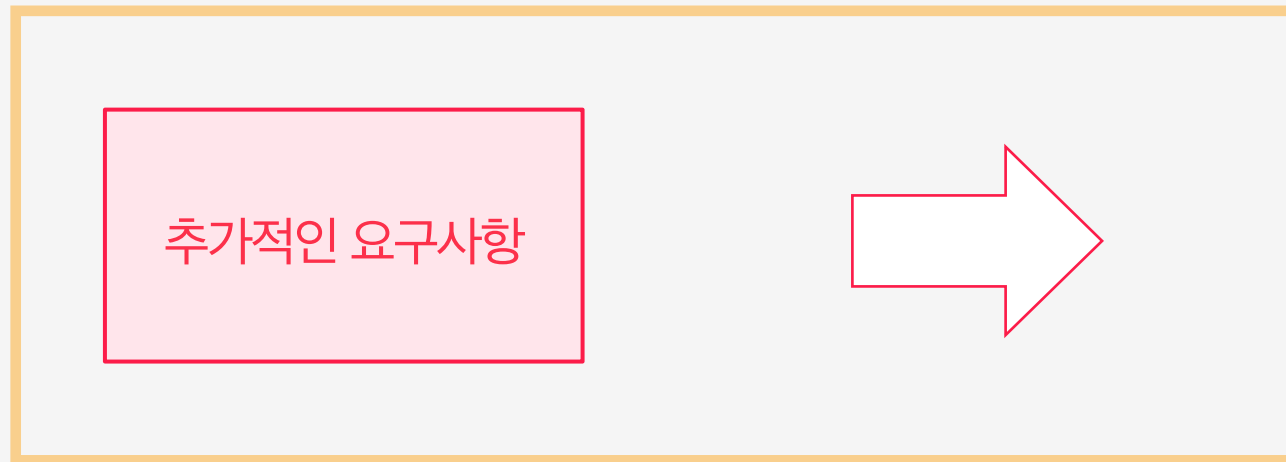
딥러닝 코딩 테스트

딥러닝 관련 과제를 1~2주 동안 해결하는 테스트

- 딥러닝 코딩 테스트 진행 시 보통 자원/시간이 충분치 않기 때문에 해보고 싶은 모든 방법론들을 다 시도하지는 못 합니다
 - 이 때 주어진 자원/시간에 따라 실험 설계를 잘 하는지 보여줄 수 있음
 - TODO 또한 잘 정리해서 결과 보고에 포함시키면 좋음

기술 면접

추상적인 요구를 구체적으로 환원시킬 수 있는지 판단하는 단계



[3]

[3] https://en.wikipedia.org/wiki/Scientific_method

기술 면접

Open-ended question에 익숙해져야 한다

- 답이 정해져 있는 질문들 보다는 특정 상황을 가정하고, 해당 상황에서 어떻게 문제를 해결할 수 있는지 묻는다
 - ex) 저희가 추천시스템 모델을 납품하는데요, 오프라인에서 모델을 평가했을 때와 온라인에서 모델을 평가 했을 때의 결과가 크게 차이가 나요. 이러한 상황을 어떻게 해결해나갈 것인가요?

일반적인 면접 준비 팁

계란을 한 바구니에 담지 마세요

- 진짜 가고 싶은 회사가 딱 하나밖에 없다고 그 회사에만 지원하지 마세요
 - 다른 회사들도 미리 지원해서 면접 경험치를 쌓고 진짜 가고 싶은 회사에 지원하는게 좋습니다

일반적인 면접 준비 팁

계란을 한 바구니에 담지 마세요

- 합격 후 연봉 협상을 진행할 경우 많은 회사에서의 오퍼가 있어야지 유리한 입장을 점할 수 있습니다
 - ex) 10개의 회사에 동시 지원 -> 4~5개의 오퍼 확보 후 연봉 협상 진행

일반적인 면접 준비 팁

면접 전에 회사에 대한 사전 조사는 필수

- ex) 리걸 테크 회사에 지원한다면 본인의 경력과 해당 회사의 제품 및 기술을 연관시킬 수 있는 내용을 미리 생각해두면 좋음
 - 왜 우리 회사에 지원 했어? 라는 질문에 대한 좋은 답변이 됨

일반적인 면접 준비 팁

모의 면접 기회가 있으면 적극 활용하세요

- 면접도 많이 해본 사람이 잘 합니다
 - 다만, 실전에서 면접을 보면서 경험치를 쌓으면 해당 면접이 본인의 평판에 영향을 끼칠 수 있다는 리스크가 있습니다
- 모의 면접은 잘 하든 못 하든 별다른 리스크 없이 면접 경험을 쌓을 수 있는 좋은 기회입니다
 - + 모의 면접은 목적 자체가 면접 스킬의 향상이어서 피드백을 받을 수 있다는 장점도 있습니다

AI Engineer

취업 후 지속가능한 성장

AI Engineer 취업 그 이후

AI 분야가 빠르게 바뀌기 때문에 취업 이후에도 계속 공부해야 합니다

- ex) 취업할 때는 추천시스템이 주 업무였지만 시간이 지나 LLM으로 업무 영역이 바뀜
- 최근 AI 분야를 강타한 LLM만 보더라도 2~3년 전에는 이걸 주로 다루는 회사가 극소수였으나 최근에는 폭증하고 있는 추세
- LLM이 나왔으니 앞으로 이것만 하면 될거라는 보장 또한 없다

AI Engineer 취업 그 이후

계속 새로운 분야를 공부하는게 너무 힘들지 않나요?

- 매번 새로운 분야를 접한다고 생각하면 어려울 수 있습니다
- 다만, 현대의 AI는 대부분 딥러닝 기반이기에 근본적으로 공통점이 아주 많습니다
 - 기본기를 착실하게 다져놓고 내가 원래 알던거에서의 변형이라고 받아들이면 생각보다 쉽게 trend를 따라갈 수 있습니다

AI Engineer 취업 그 이후

딥러닝의 근간

- 매번 새로운 분야를 접한다고 생각하면 어려울 수 있습니다
- 다만, 현대의 AI는 대부분 딥러닝 기반이기에 근본적으로 공통점이 아주 많습니다
 - Data-driven 사고 방식
 - 데이터 전처리
 - 데이터 증강
 - 최적화 기법

AI Trend 따라가기

Linkedin 등 비즈니스 SNS 활용하기

- Linkedin 처럼 비즈니스 성향이 강한 SNS에는 최근 분야에서 핫한 아이디어들이 활발히 공유됩니다
 - Linkedin을 통해서 네트워킹도 하고, 분야의 최근 trend를 적절한 거리를 두고 파악하고 있기에 용이

AI Trend 따라가기

사내 모임 활성화 하기

- Reading group, paper seminar 등 이름은 다양하지만 결국 최신 페이퍼를 읽고 논의하는 자리
 - 보통 일주일에 특정 시간을 지정하고 다같이 참여
 - 롱런하기 위해서는 운영을 잘 해야됨

Thank you :)

